# Software Requirements Specification

for

# E Crime File Management

Version 1.0

Prepared by

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May 16, 2015

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# Revision History

| Name | Date | Reason For Changes | Version |
|---|---|---|---|
| | | | |
| | | | |

# 1. Introduction

This document is a software requirement specification for the "E Crime File Management System" for web and android.

## 1.1 Purpose

[Specify the purpose of this **SRS**. The **SRS** should fully describe the external behavior of the application or subsystem identified. It also describes nonfunctional requirements, design constraints and other factors necessary to provide a complete and comprehensive description of the requirements for the software.]

## 1.2 Scope

The proposed project is purely focused to help the people in registering first information reports (FIR) and help them monitor their case's progress without the need of visiting the police station. The project will not allow the police stations to keep record of their employees. The project is divided into different sections the first section will handle registration of First Information Report and reporting a missing person, whereas the second section is focused on managing records of the police stations, prisoners and the most wanted criminals along with the information crimes. The final section will allow people to see police case diaries which will be updated by the concerned police officer.

## Web application

- Report a Crime
  - File a F.I.R.
  - Report a missing person
  - Update F.I.R
  - Monitor progress of the case.
  - List of evidences and witnesses related to each case
- Police station Record Management
  - Add a new police station
  - Updating information of police station
  - Maintain a list of missing persons
  - Keeping information of prisoners

- Criminal Record Management

  - Adding new criminals to database
  - Updating status of criminals
  - Updating and deleting a police station from the system
  - View and reply user complaint
  - Updating list of most wanted criminals

- User Management

**Police officer:**

- Login In
- Log Out
- Manage records of criminals
- Updating each case progress
- Updating information of police station
- Update information about prisoners

**Citizen user:**

- Log In
- Log Out
- Monitor their complaint
- File a complaint
- View crimes history of month wise
- Registered user can send messages and receive messages from administrator.
- Contact police station via personal message

**Administrator:**

- Access all data
- Populate the database
- Manage the users
- Add new updates of police officers
- Add new admin
- Add latest news of recent successfully completed cases
- Admin to chat with the administrator or with other registered users

- Reporting
  - Chart and graphs based crime analysis reports
  - Crime category wise reports

## Android application

- Report a Crime
  - File a F.I.R.
  - Report a missing person
  - Update F.I.R
  - Monitor progress of the case
  - List of evidences and witnesses related to each case
- User Account Management
  - Log In
  - Log Out
  - Monitor their complaint
  - File a complaint
  - View crimes reports month wise
  - Registered user can send messages and receive messages from administrator.
  - Contact police station via personal message
  - View criminal's data by searching

## 1.3 Definitions, Acronyms and Abbreviations

[This subsection should provide the definitions of all terms, acronyms, and abbreviations required to properly interpret the **SRS**. This information may be provided by reference to the project Glossary.]

## 1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- Roger S. Pressman, Software Engineering, A Practitioner Approach 7th edition 2010.

## 1.5 Overview

# 2. Overall Description

[This section of the **SRS** should describe the general factors that affect the product and its requirements. This section does not state specific requirements. Instead, it provides a background for those requirements, which are defined in detail in Section 3, and makes them easier to understand. Include such items as:

- product perspective
- product functions
- user characteristics
- constraints
- assumptions and dependencies
- requirements subsets]

# 3. Use Case(s)

| | |
|---|---|
| **Use Case ID:** | UC_01 |
| **Use Case Name:** | Login Account |
| **Actors:** | User/Admin/Police Officer |
| **Description:** | User selects log in option and provides required credentials and is redirected to home page. |
| **Trigger:** | User clicks the "login" button to get access to the system. |
| **Preconditions:** | User is already registered to E Crime File Management System. |
| **Postconditions:** | The system redirects the user to the home page. |
| **Normal Flow:** | 1. The user goes to the welcome page of E Crime File System by entering the URL.<br>2. User clicks login option.<br>3. The system prompts the user to enter valid username and password.<br>4. The user enters required credentials and presses the login button present on the login form.<br>5. The system verifies the given information of the user and if it is correct, the system redirects the user to the home page. |
| **Alternative Flows:** **[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | 4a. In step 4 of normal flow. If user enters incorrect username or password or both.<br>1) Log in is disapproved.<br>2) The system displays the message to re-enter the username or password.<br>3) The user enters correct username or password or both.<br>4) Use case resumes on step 3. |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | User knows the username and password. |
| **Notes and Issues:** | The maximum size of the username will be consisting of not more than 15 characters and password will be consisting of 8 characters. |

| | |
|---|---|
| **Use Case ID:** | UC_02 |
| **Use Case Name:** | Sign up |
| **Actors:** | User |
| **Description:** | User selects sign up option and system shows the sign up form to the user. |
| **Trigger:** | User clicks the "sign up" button to create an account. |
| **Preconditions:** | None |
| **Postconditions:** | New user account is created. |
| **Normal Flow:** | 1. The user goes to the welcome page of E Crime File System by entering the URL.<br>2. User clicks sign up option.<br>3. The system prompts the user to enter valid username, password, first name, last name, CNIC number etc.<br>4. The user enters required credentials and presses the sign up button present on the login form.<br>5. The system verifies that given information of the user doesn't already exist in the system and if the information provided by the user is correct, new account is created for the user. |
| **Alternative Flows:** **[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | 5a. In step 5 of normal flow. If user account with same information already exists in the system.<br>1) Sign up operation is unsuccessful.<br>2) The system displays the message to enter another unique username.<br>3) The user enters another unique username.<br>4) Use case resumes on step 4. |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | The username must be unique. The maximum size of the username will be consisting of not more than 15 characters and password will be consisting of 8 characters. |

| | |
|---|---|
| **Use Case ID:** | UC_03 |
| **Use Case Name:** | Report Missing Person |
| **Actors:** | User |
| **Description:** | User reports to the police that a person is missing. |
| **Trigger:** | User clicks the "report missing person" option to report the case. |
| **Preconditions:** | User is already logged in. |
| **Postconditions:** | Complaint is sent to police officer. |
| **Normal Flow:** | 1. The user clicks report missing person option.<br>2. The system displays report missing person's complaint page.<br>3. The user enters details about the person who is missing.<br>4. The user submits the complaint and system sends reported case to the concerning police officer's account.<br>5. The feedback message is displayed to the user. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | 4a. In step 4 of normal flow. If user doesn't select concerning police station.<br><br>1. The system will prompt the user to select a police station.<br>2. The user will select the concerning police station.<br>3. The use case resumes at step 4. |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_04 |

| **Use Case Name:** | Report FIR |
|---|---|
| **Actors:** | User |
| **Description:** | User reports to the police about a crime incident. |
| **Trigger:** | User clicks the "report FIR" option to report the any criminal activity. |
| **Preconditions:** | User is already logged in. |
| **Postconditions:** | FIR is sent to police officer for review and action. |
| **Normal Flow:** | 1. The user clicks report FIR.<br>2. The system displays first information report's page to the user.<br>3. The user enters details about the crime incident.<br>4. The user submits the FIR.<br>5. The system verifies user's given information like police station selected by user and sends reported case to the concerning police officer's account.<br>6. The feedback message is displayed to the user. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | 4a. In step 4 of normal flow. If user doesn't select concerning police station.<br><br>1. The system will prompt the user to select a police station.<br>2. The user will select the concerning police station.<br>3. The use case resumes at step 4. |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| **Use Case ID:** | UC_05 |
|---|---|
| **Use Case Name:** | Monitor case's progress |

| | |
|---|---|
| **Actors:** | User |
| **Description:** | User checks for the progress made in the investigation phase of the criminal case reported by that particular user. |
| **Trigger:** | User clicks the "view case progress" option to view progress made in investigation. |
| **Preconditions:** | User is already logged in and at least one case is registered by the user. |
| **Postconditions:** | Complete case progress |
| **Normal Flow:** | 1. The user clicks view case progress option.<br>2. The system displays list of all cases registered by that particular user.<br>3. The user selects one of the cases from given list.<br>4. The system redirects the user to that case's page with complete information of the progress made in that case. |
| **Alternative Flows:** **[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | There must be a case registered by the user before, he/she can monitor the progress of that case. |

| | |
|---|---|
| **Use Case ID:** | UC_06 |
| **Use Case Name:** | Add new evidences |
| | |
| **Actors:** | User |
| **Description:** | User adds evidences to a particular case which has already been |

| | |
|---|---|
| | registered by that user. |
| **Trigger:** | User clicks “add new evidence” option. |
| **Preconditions:** | User is already logged in and at least one case is registered by the user. |
| **Postconditions:** | New evidences added to the case |
| **Normal Flow:** | 1. The user clicks view case progress option.<br>2. The system displays list of all cases registered by that particular user.<br>3. The user clicks one of the cases from given list.<br>4. The system redirects the user to that case’s page with complete information of the progress made in that case.<br>5. Then user clicks on add new evidences option.<br>6. User selects files to be attached as evidences.<br>7. User submits the evidences and the system sends case file to the police officer. |
| **Alternative Flows:** **[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | 4a. In step 4 of normal flow. If user doesn’t select evidence file.<br><br>1. The system will prompt the user to select a file to be attached as evidence.<br>2. The user will select the concerning police station.<br>3. The use case resumes at step 4. |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | There must be a case registered by the user before, he/she can add supplementing evidences. |

| | |
|---|---|
| **Use Case ID:** | UC_07 |
| **Use Case Name:** | Update user account |
| **Actors:** | User |
| **Description:** | User updates his/her personal information. |
| **Trigger:** | User clicks “update profile” option. |

| | |
|---|---|
| **Preconditions:** | User is already logged in. |
| **Postconditions:** | User profile is updated. |
| **Normal Flow:** | 1. The user clicks update profile option.<br>2. The system redirects the user to update profile page.<br>3. The user modifies existing information.<br>4. The user submits updated profile information.<br>5. The system verifies that username remains unique and updates the user's account.<br>6. The system displays the feedback message to the user. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | 5a. In step 5 of normal flow. If user changes username.<br>1. System will display message error message and prompt the user to enter unique username.<br>2. The use case resumes at step 3. |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | The username of the user must remain unique. |

| | |
|---|---|
| **Use Case ID:** | UC_08 |
| **Use Case Name:** | Contact Police Officer |
| **Actors:** | Primary: User<br>Secondary: System |
| **Description:** | User sends message to police officer. |
| **Trigger:** | User clicks "contact police officer" button. |
| **Preconditions:** | User is already logged in and user has already registered a case. |
| **Postconditions:** | User's message is sent to contact police officer. |
| **Normal Flow:** | 1. The user clicks view case progress option.<br>2. The system displays list of all cases registered by that particular user.<br>3. The user clicks one of the cases from given list.<br>4. The system redirects the user to that case's page with complete information of the progress made in that case.<br>5. Then user clicks on contact police officer button.<br>6. User enters the desired message and sends it.<br>7. The system transfers the message to police officer's account. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_09 |
| **Use Case Name:** | Update FIR |
| **Actors:** | Primary: User<br>Secondary: Police Officer |
| **Description:** | User updates the incident information of reported FIR. |
| **Trigger:** | User clicks "update FIR" option. |
| **Preconditions:** | User is already logged in and user has already registered a case. |
| **Postconditions:** | FIR is sent to police officer for review. |
| **Normal Flow:** | 1. The user clicks view case progress option.<br>2. The system displays list of all cases registered by that particular user.<br>3. The user clicks one of the cases from given list.<br>4. The system redirects the user to that case's page with complete information of the progress made in that case.<br>5. Then user clicks on update FIR option.<br>6. User updates the given information and submits the update FIR.<br>7. The system transfers the FIR to police officer's account for a review. |
| **Alternative Flows:** **[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | 6a. In step 6 of normal flow. If the user leaves incident details section empty.<br>1. System will display message error message and prompt the user to enter incident details.<br>2. The use case resumes at step 6. |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | The FIR will only be updated after the police officer approves it. |

| | |
|---|---|
| **Use Case ID:** | UC_10 |
| **Use Case Name:** | Contact Admin |
| **Actors:** | Primary: User<br>Secondary: System |
| **Description:** | User sends message to the administrator. |
| **Trigger:** | User clicks “Contact admin” option. |
| **Preconditions:** | User is already logged in. |
| **Postconditions:** | Message is sent to the admin. |
| **Normal Flow:** | 1. The user clicks contact admin option.<br>2. The system displays the contact admin page.<br>3. The user enters message for the admin.<br>4. The system sends message to the admin. |
| **Alternative Flows:**<br>**Flow 1 – Not in Network]** | None<br>**[Alternative** |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_11 |
| **Use Case Name:** | Receive Messages |
| **Actors:** | User |
| **Description:** | User receives messages from admin/police officer. |
| **Trigger:** | User clicks “view messages” option. |
| **Preconditions:** | User is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The user clicks view messages option.<br>2. The system redirects the user to the messages page.<br>3. The user clicks a particular message to view it. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_12 |
| **Use Case Name:** | View most wanted criminals |
| **Actors:** | User |
| **Description:** | User views details of most wanted criminal. |
| **Trigger:** | User clicks "most wanted criminals" option. |
| **Preconditions:** | None |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The user clicks most wanted criminals option.<br>2. The system displays list of all criminals that exist in the system's records.<br>3. The user clicks one of the criminals from given list.<br>4. The system redirects the user to that criminal's profile page with complete information of that criminal. |
| **Alternative Flows:** **[Alternative Flow 1 – Not in Network]** | 1. The user enters the name of a criminal in the search bar.<br>2. The system displays list of criminals that match the given name.<br>3. The user clicks the desired criminal and is redirected to the page of that particular criminal's profile page. |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | The criminal must exist in the system records in order to be searched. |

| | |
|---|---|
| **Use Case ID:** | UC_13 |
| **Use Case Name:** | View missing persons |
| **Actors:** | User |
| **Description:** | User views details of missing person. |
| **Trigger:** | User clicks “missing persons” option. |
| **Preconditions:** | None |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The user clicks missing person option.<br>2. The system displays list of all missing persons that exist in the system’s records.<br>3. The user clicks one of the missing persons from given list.<br>4. The system redirects the user to that missing person’s profile page with complete information of that person. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | 1. The user enters the name of a missing person in the search bar.<br>2. The system displays list of missing persons that match the given name.<br>3. The user clicks the desired person and is redirected to the page of that particular missing person’s profile page. |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | The missing person must exist in the system records in order to be searched. |

| | |
|---|---|
| **Use Case ID:** | UC_14 |
| **Use Case Name:** | Logout |
| **Actors:** | User/admin/police officer |
| **Description:** | User logs out of his/her account. |
| **Trigger:** | User clicks “log out” button. |
| **Preconditions:** | None |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The user clicks log out button.<br>2. The system redirects the user to the welcome page. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_15 |
| **Use Case Name:** | Review user's complaint |
| **Actors:** | Police Officer |
| **Description:** | Police Officer views details of criminal case reported. |
| **Trigger:** | Police Officer clicks "Reported cases" option. |
| **Preconditions:** | Police officer is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The police officer clicks reported cases option.<br>2. The police officer clicks one of the listed cases.<br>3. The police officer reviews the information given the by the user.<br>4. The police officer updates the status of the case. |
| **Alternative Flows:** **[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_16 |
| **Use Case Name:** | Update case progress |
| **Actors:** | Police Officer |
| **Description:** | Police Officer updates the progress made in investigation process. |
| **Trigger:** | Police Officer clicks "Update progress" button. |
| **Preconditions:** | Police officer is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The police officer clicks reported cases option.<br>2. The police officer clicks one of the listed cases.<br>3. The police officer reviews the information given the by the user.<br>4. The police officer updates the case investigation progress by adding the witnesses interviewed and evidences found.<br>5. The Police Officer saves the progress. |
| **Alternative Flows:**<br><br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_17 |
| **Use Case Name:** | Respond Messages |
| **Actors:** | Police Officer/Admin |
| **Description:** | User views and replies user messages. |
| **Trigger:** | User clicks "reply" button. |
| **Preconditions:** | User is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The user clicks messages.<br>2. The user selects a message and opens it.<br>3. The user clicks button and types messages.<br>4. The user clicks send button. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_18 |
| **Use Case Name:** | Delete complaint |
| **Actors:** | Police Officer |
| **Description:** | Police Officer deletes the complaint. |
| **Trigger:** | Police Officer clicks "delete" button. |
| **Preconditions:** | Police officer is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The police officer clicks reported cases option.<br>2. The police officer clicks one of the listed cases.<br>3. The police officer reviews the information given the by the user.<br>4. The police officer clicks delete button if it is a fake complaint or a spam. |
| **Alternative Flows:**<br><br>**Flow 1 [Alternative Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_19 |
| **Use Case Name:** | View Crime Statistics |
| **Actors:** | User |
| **Description:** | User views the crime statistics. |
| **Trigger:** | User clicks "crime statistics" option. |
| **Preconditions:** | None |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The user clicks crime statistics option.<br>2. The user is redirected to crime statistics page. |
| **Alternative Flows:** **[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_20 |
| **Use Case Name:** | Add new police station |
| **Actors:** | Admin |
| **Description:** | Admin adds information of new police station to the system. |
| **Trigger:** | Admin clicks "add police station" option. |
| **Preconditions:** | Admin is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The admin clicks add new police station option.<br>2. The admin enters information of the new police station.<br>3. The admin clicks submit button and new police station is added. |
| **Alternative Flows:** **[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_21 |
| **Use Case Name:** | Delete police station |
| **Actors:** | Admin |
| **Description:** | Admin deletes information of new police station to the system. |
| **Trigger:** | Admin clicks “delete police station” button. |
| **Preconditions:** | Admin is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The admin selects the desired police station.<br>2. The admin clicks the delete police station button.. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_22 |
| **Use Case Name:** | Publish missing persons list |
| **Actors:** | Admin |
| **Description:** | Admin publishes list of missing persons. |
| **Trigger:** | Admin clicks "publish list" button. |
| **Preconditions:** | Admin is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The admin clicks view missing person option.<br>2. The admin updates the current missing persons.<br>3. The admin clicks publish list button. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_23 |
| **Use Case Name:** | Add prisoner |
| **Actors:** | Admin |
| **Description:** | Admin adds new prisoner to the police station. |
| **Trigger:** | Admin clicks "add prisoner" option. |
| **Preconditions:** | Admin is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The admin clicks add prisoner option.<br>2. The admin selects police station and adds information of the prisoner.<br>3. The admin clicks add prisoner button. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_24 |
| **Use Case Name:** | Add most wanted criminal |
| **Actors:** | Admin |
| **Description:** | Admin adds most wanted criminal to the system. |
| **Trigger:** | Admin clicks “add criminal” option. |
| **Preconditions:** | Admin is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The admin clicks add most wanted criminal option.<br>2. The admin enters details of the criminal.<br>3. The admin clicks submit button and record is saved in the system. |
| **Alternative Flows:**<br><br>**Flow 1 [Alternative Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_25 |
| **Use Case Name:** | Update most wanted criminals |
| | |
| **Actors:** | Admin |
| **Description:** | Admin updates information of the most wanted criminal. |
| **Trigger:** | Admin clicks "update criminal" button. |
| **Preconditions:** | Admin is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The admin clicks view most wanted criminals.<br>2. The admin clicks the desired criminal link.<br>3. The admin clicks update criminal button.<br>4. The admin enter new details of the criminal.<br>5. The admin clicks the submit button. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_26 |
| **Use Case Name:** | Delete most wanted criminals |
| **Actors:** | Admin |
| **Description:** | Admin deletes most wanted criminal. |
| **Trigger:** | Admin clicks "delete" button. |
| **Preconditions:** | Admin is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The admin clicks view most wanted criminals.<br>2. The admin clicks the desired criminal link.<br>3. The admin clicks delete criminal button. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

| | |
|---|---|
| **Use Case ID:** | UC_27 |
| **Use Case Name:** | Update crime statistics |
| **Actors:** | Admin |
| **Description:** | Admin updates crime statistics. |
| **Trigger:** | Admin clicks "update statistics" button. |
| **Preconditions:** | Admin is already logged in. |
| **Postconditions:** | None |
| **Normal Flow:** | 1. The admin clicks add crime statistics option.<br>2. The admin updates the crime statistics.<br>3. The admin clicks update statistics button. |
| **Alternative Flows:**<br>**[Alternative Flow 1 – Not in Network]** | None |
| **Exceptions:** | None |
| **Includes:** | None |
| **Special Requirements:** | The system shall respond the user within one second. |
| **Assumptions:** | None |
| **Notes and Issues:** | None |

# 4. Specific Requirements

## 4.1 Functionality

Major functionalities of E Crime File Management are:

### 4.1.1 Sign Up

| Identifier | FR1 |
|---|---|
| Title | Sign Up |
| Requirement | The system shall provide an interface to the user to create an account to access the utilities provided by the system by providing information e.g. First name, Last name, CNIC number etc. |
| Source | Team Members |
| Rationale | To create user account |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

### 4.1.2 Admin and Police Login

| Identifier | FR2 |
|---|---|
| Title | Admin and Police Login |
| Requirement | The system shall allow the administrator to get access to the system by providing username and password. |
| Source | Team Members |
| Rationale | To access the application and manage the user requests. |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

### 4.1.3 User Login

| Identifier | FR3 |
|---|---|
| Title | User Login |
| Requirement | The system shall allow the user to get access to the system by providing username and password. |
| Source | Team Members |
| Rationale | To access the application and to utilize its features after logging successfully. |
| Restrictions and Risk | None |
| Dependencies | FR1 |
| Priority | High |

### 4.1.4 Logout (admin and user)

| Identifier | FR4 |
|---|---|
| Title | Logout |
| Requirement | The system shall provide logout interface for admin and users to logout. |
| Source | Team Members |
| Rationale | For secure use of the system. |
| Restrictions and Risk | None |
| Dependencies | FR1,FR2,FR3 |
| Priority | High |

### 4.1.5 Update Profile

| Identifier | FR5 |
|---|---|
| Title | Update profile |
| Requirement | The system shall provide the feature to allow users to update their personal information after their account is created. |
| Source | Team members. |
| Rationale | To keep users information consistent. |
| Restrictions and Risk | None |
| Dependencies | FR3 |
| Priority | Medium |

### 4.1.6 Report Missing Person

| Identifier | FR6 |
|---|---|
| Title | Report missing person |
| Requirement | The system shall provide the feature to the users to report any missing person to the police. |
| Source | Team Members |
| | |
| Rationale | Allow users to report to police. |
| Restrictions and Risk | None |
| Dependencies | FR3 |
| Priority | High |

### 4.1.7 Register FIR

| Identifier | FR7 |
|---|---|
| Title | Report FIR |
| Requirement | The system shall provide the interface to the user to report any criminal activity. |
| Source | Team Members |
| Rationale | To allow users register criminal complaints. |
| Restrictions and Risk | None |
| Dependencies | FR3 |
| Priority | High |

### 4.1.8 Contact Police

| Identifier | FR8 |
|---|---|
| Title | Contact Police |
| Requirement | The system shall provide the feature to allow the user to contact the police officer. |
| Source | Team Members |
| Rationale | User might want to discuss the case with police. |
| Restrictions and Risk | None |
| Dependencies | FR3,FR7 |
| Priority | Medium |

### 4.1.9 View Case Progress

| Identifier | FR9 |
|---|---|
| Title | View Case Progress |
| Requirement | The system shall provide the interface to the user to monitor the progress made in the investigation process of the case. |
| Source | Team Members |
| Rationale | To allow users monitor the case progress. |
| Restrictions and Risk | None |
| Dependencies | FR3 |
| Priority | |

### 4.1.10 Update FIR

| Identifier | FR10 |
|---|---|
| Title | Update FIR |
| Requirement | The system shall provide the interface to the user to make changes in the FIR reported. |
| Source | Team Members |
| Rationale | To allow the users change the details in FIR. |
| Restrictions and Risk | None |
| Dependencies | FR3,FR7 |
| Priority | Medium |

### 4.1.11 Add new evidence

| Identifier | FR11 |
|---|---|
| Title | Add New Evidence |
| Requirement | The system shall allow the interface to add new evidence to the reported case. |
| Source | Team Members |
| Rationale | To allow users add more evidences to the reported FIR. |
| Restrictions and Risk | None |
| Dependencies | FR3,FR7 |
| Priority | Medium |

### 4.1.12 Contact Admin

| Identifier | FR12 |
|---|---|
| Title | Contact Admin |
| Requirement | The system shall provide the feature to the user to contact the admin. |
| Source | Team Members |
| | |
| Rationale | To allow users contact admin. |
| Restrictions and Risk | None |
| Dependencies | FR3 |
| Priority | |

### 4.1.13 View Wanted Criminals

| Identifier | FR13 |
|---|---|
| Title | View wanted Criminals |
| Requirement | The system shall allow the user to view the details of most wanted criminals. |
| Source | Team Members |
| Rationale | Users might be interested in viewing most wanted criminals. |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

### 4.1.14 Check Crime Statistics

| Identifier | FR14 |
|---|---|
| Title | Check Crime Statistics |
| Requirement | The system shall provide the interface in the form of graphs and charts to show crime statistics to the user. |
| Source | Team Members |
| Rationale | To allow users check crime statistics. |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

### 4.1.15 Add Criminal

| Identifier | FR15 |
|---|---|
| Title | Add Criminal |
| Requirement | The system shall provide the interface to admin to add new criminals to the system. |
| Source | Team Members |
| | |
| Rationale | To allow admin add new criminals. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | High |

### 4.1.16 Delete Criminal

| Identifier | FR16 |
|---|---|
| Title | Delete Criminal |
| Requirement | The system shall provide the interface to admin to delete criminals from the system. |
| Source | Team Members |
| Rationale | To allow admin delete criminals. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | High |

### 4.1.17 Update Criminal

| Identifier | FR17 |
|---|---|
| Title | Update Criminal |
| Requirement | The system shall provide the interface to admin to update criminal information to the system. |
| Source | Team Members |
| Rationale | To allow admin update criminals. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | High |

### 4.1.18 Update Crime Statistics

| Identifier | FR18 |
|---|---|
| Title | Update Crime Statistics |
| Requirement | The system shall provide the interface to admin to update criminal crime statistics. |
| Source | Team Members |
| Rationale | Admin shall be able to update crime statistics. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | High |

### 4.1.19 Add Police Station

| Identifier | FR19 |
|---|---|
| Title | Add police station |
| Requirement | The system shall allow the admin to add new police stations to the system. |
| Source | Team Members |
| Rationale | Admin shall be able to add new police station. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | High |

### 4.1.20 Update Case Progress

| Identifier | FR20 |
|---|---|
| Title | Update Case Progress |
| Requirement | The system shall provide the feature to allow police officer to update progress of investigation. |
| Source | Team Members |
| Rationale | Police officer shall be able to update case progress. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | High |

### 4.1.21 Add Prisoner

| Identifier | FR21 |
|---|---|
| Title | Add Prisoner |
| Requirement | The system shall provide the feature of adding new prisoners to a police station. |
| Source | Team Members |
| Rationale | Admin shall be able to add police station. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | High |

### 4.1.22 Publish Missing Persons

| Identifier | FR22 |
|---|---|
| Title | Publish Missing Person |
| Requirement | The system shall let admin generate list of missing persons. |
| Source | Team Members |
| Rationale | Admins shall be able to publish missing persons. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | High |

### 4.1.23 Receive Messages

| Identifier | FR23 |
|---|---|
| Title | Receive Messages |
| Requirement | The system shall provide the feature of receiving messages for user. |
| Source | Team Members |
| Rationale | User shall be able to receive messages. |
| Restrictions and Risk | None |
| Dependencies | FR3 |
| Priority | High |

### 4.1.24 Update Missing Persons

| Identifier | FR24 |
|---|---|
| Title | Update Missing Persons |
| Requirement | The system shall allow the admin to update status of missing persons. |
| Source | Team Members |
| Rationale | Admin shall be able to update missing person. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | Medium |

### 4.1.25 Update Police Station

| Identifier | FR25 |
|---|---|
| Title | Update Police Station |
| Requirement | The system shall provide the feature to update information of police stations. |
| Source | None |
| Rationale | Admin shall be able to update police station details. |
| Restrictions and Risk | None |
| Dependencies | FR2 |
| Priority | Medium |

## 4.2 Usability

### 4.2.1 Understandability

| Identifier | UR1 |
|---|---|
| Title | Understandability |
| Requirement | The user interface shall be easy to use; all the user interface elements shall be self-explanatory. |
| Source | Team Members |
| Rationale | None |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

### 4.2.2 Learning Time

| Identifier | UR2 |
|---|---|
| Title | Learning Time |
| Requirement | The system shall provide the GUI that any normal user of windows could learn it in 15 minutes and the Other user (i.e. not user of windows or any other Operating System) could learn in 1 hour maximum. |
| Source | Team Members |
| Rationale | None |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

## 4.3 Reliability

### 4.3.1 Availability

| Identifier | RR1 |
|---|---|
| Title | Availability |
| Requirement | The system shall provide 24/7 services to the user. |
| Source | Team Members |
| Rationale | None |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

## 4.4 Performance

### 4.4.1 Number of Users

| Identifier | PR1 |
|---|---|
| Title | Number Of Users |
| Requirement | There is no restriction on number of users that can use the system in parallel manner. |
| Source | Team Members |
| Rationale | None |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

### 4.4.2 Compatibility

| Identifier | PR2 |
|---|---|
| Title | Compatibility |
| Requirement | The system shall be compatible with all the major browsers and android app shall run on minimum of 2.3 version of android operating system. |
| Source | Team Members |
| Rationale | None |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

## 4.5 Supportability

### 4.5.1 Naming Conventions

| Identifier | SR1 |
|---|---|
| Title | Naming Conventions |
| Requirement | Standard naming conventions of smartphones will be used in developing the project, such as image gallery, video gallery, messages inbox etc. |
| Source | Team Members |
| Rationale | None |
| Restrictions and Risk | None |
| Dependencies | None |
| Priority | High |

## 4.6 Design Constraints

### 4.6.1 Software language

The system will be developed in java, HTML5, CSS3, JQuery, Javascript.

### 4.6.2 Development Tools

Any version of eclipse will be used as development tool.

## 4.7 On-line User Documentation and Help System Requirements

There will be no need of online user documentation, as the system shall be self-explanatory itself.

## 4.8 Purchased Components

Domain for running our website officially.

## 4.9 Interfaces

### 4.9.1 User Interfaces

[Describe the user interfaces that are to be implemented by the software.]

### 4.9.2 Hardware Interfaces

[This section defines any hardware interfaces that are to be supported by the software, including logical structure, physical addresses, expected behavior, etc. ]

### 4.9.3 Software Interfaces

[This section describes software interfaces to other components of the software system. These may be purchased components, components reused from another application or components being developed for subsystems outside of the scope of this **SRS** but with which this software application must interact.]

### 4.9.4 Communications Interfaces

- E Crime File Management System will be connected to the World Wide Web.
- The HTTPS protocol will be used to facilitate communication between the client and server.
- It will use port 8080.

### 4.10 Licensing Requirements

Not applicable.

### 4.11 Legal, Copyright, and Other Notices

E Crime File Management System will be created and assembled by students of CIIT Shahrukh and Nadeem Mughal. So, copying this or any changes to the software without permission of them and CIIT will not be tolerated. All legal rights belong to them.

### 4.12 Applicable Standards

Quality of our system will be important issue, which will be kept in mind while development. Easy design and better error handling and recovery technique will be the factors in quality of software. Application standards will be approximately same as kept in similarly developed software's.

## 5. Supporting Information

[The supporting information makes the **SRS** easier to use. It includes:

- Table of contents
- Index
- Appendices

These may include use-case storyboards or user-interface prototypes. When appendices are included, the **SRS** should explicitly state whether or not the appendices are to be considered part of the requirements.]